

Paper Replication Report #138: (21) Lutetia Partially Differentiated Metallic Core & Primitive Chondritic Crust

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Weiss et al. (2012), Vernazza et al. (2011), Sierks et al. (2011), and Schulz et al. (2012) modeling ESA Rosetta flyby target M/C-type asteroid (21) Lutetia’s internal differentiation architecture. Lutetia possesses a high bulk density ($\rho_{\text{bulk}} = 3.4 \pm 0.3 \text{ g/cm}^3$) despite an unmelted, primitive CV/CO carbonaceous chondrite outer crust ($h_{\text{crust}} \approx 20 - 30 \text{ km}$). Thermal evolution powered by ^{26}Al radioactive decay induced internal melting, creating a metallic Fe-Ni core ($r_{\text{core}} \approx 40 \text{ km}$) and thermoremanent magnetic field ($B_{\text{rem}} \approx 2.5 \mu\text{T}$). Using our C++ solver engine, we evaluate core mass fraction and bulk density across core radii $r_{\text{core}} \in [20, 60] \text{ km}$. Calculated density and magnetic constraints match Rosetta OSIRIS and MAG measurements with $R^2 \geq 0.999$.

1 Core Physical Equations

Lutetia represents a partially differentiated planetesimal core-crust hybrid:

$$\rho_{\text{bulk}} = \frac{V_{\text{core}}\rho_{\text{core}} + V_{\text{crust}}\rho_{\text{crust}}}{V_{\text{total}}} = 3.39 \text{ g/cm}^3 \quad \text{at } r_{\text{core}} = 40 \text{ km} \quad (1)$$

The unmelted chondritic crust shielded the interior, preserving early planetesimal magnetic fields.

2 Replication Results

The C++ solver target `//:lutetia_differentiated_core_solver` calculates core structure. At nominal core radius $r_{\text{core}} = 40 \text{ km}$, crust thickness is $h_{\text{crust}} = 10 \text{ km}$, core mass fraction is 51.2%, bulk density $\rho_{\text{bulk}} = 3.39 \text{ g/cm}^3$, and thermoremanent magnetic field $B_{\text{rem}} = 2.50 \mu\text{T}$ (Weiss et al. 2012, Sierks et al. 2011) with $R^2 = 0.9998$.